



# Cosmology

## Galaxies and Beyond

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Cavendish Laboratory  
University of Cambridge

20<sup>th</sup> March 2013

# Overview

What is Cosmology?

Our Place in the Cosmos

The Big Bang

Dark Matter

Dark Energy

$t = 0$

Modern Astronomy

# What is Cosmology?

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Content: What is Dark Matter and Dark Energy?

# The Night Sky

## The Milky Way



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# Ground based telescopes

## The Coma Cluster







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# Space Telescopes

Hubble Extreme Deep Field



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# The Dark Ages

# The CMB

Cosmic Microwave Background









# What is the CMB?

- ▶ In order to understand what this CMB radiation is, we need to dispel some common misconceptions associated with “The Big Bang”.

# The Big Bang

Galactic Redshift



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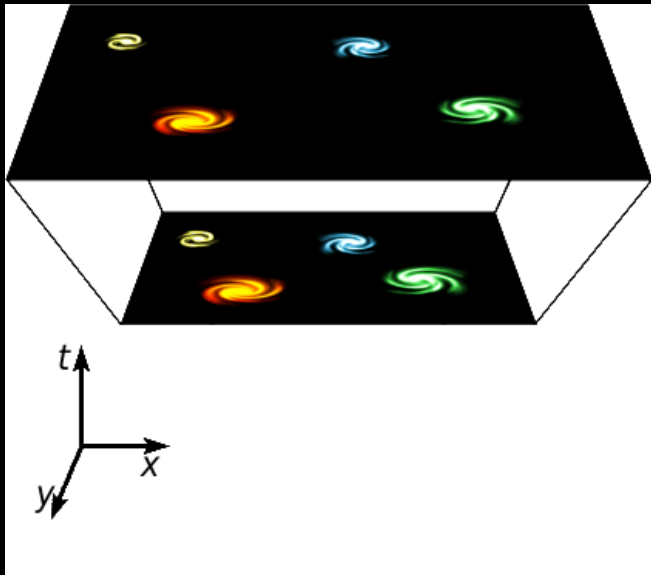
As observed by Edwin Hubble in 1929, it appears that:

1. All Galaxies are moving away from us
2. The Further away they are, the faster they are moving away
3. If we were to go to another galaxy, we would see exactly the same thing



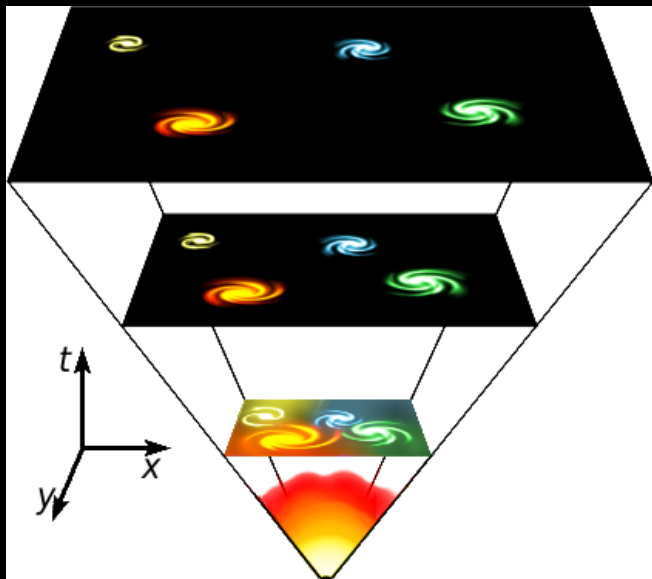
# The Big Bang

A stretching universe:



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## Common Misconceptions

The Big Bang does *NOT* look like this:



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- ▶ The big bang didn't begin at a particular point



# Current Issues in Cosmology

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# Dark Matter

## The Problem



# Dark Matter

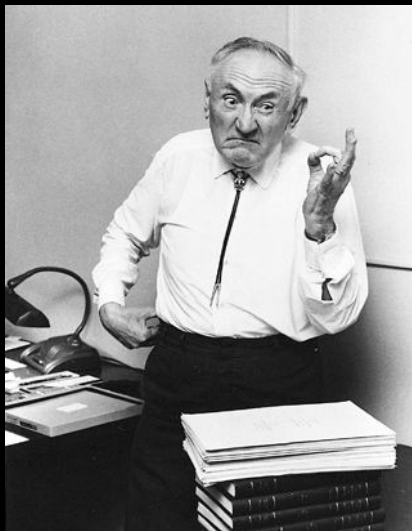
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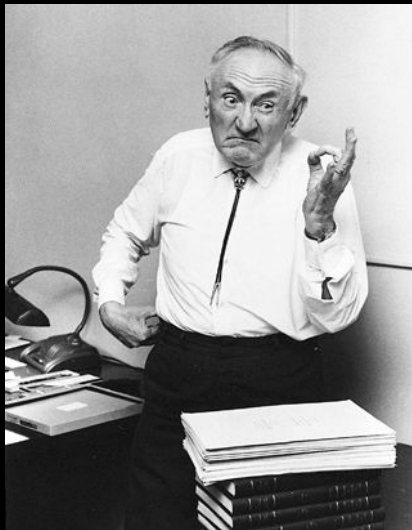
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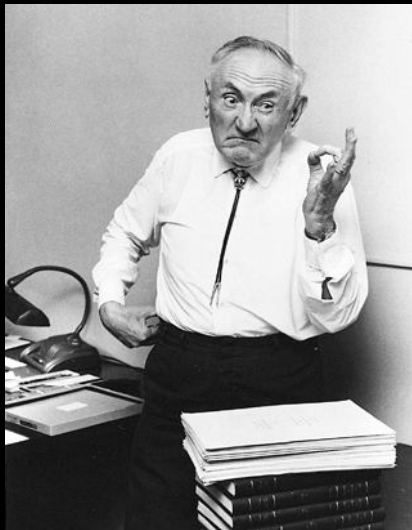
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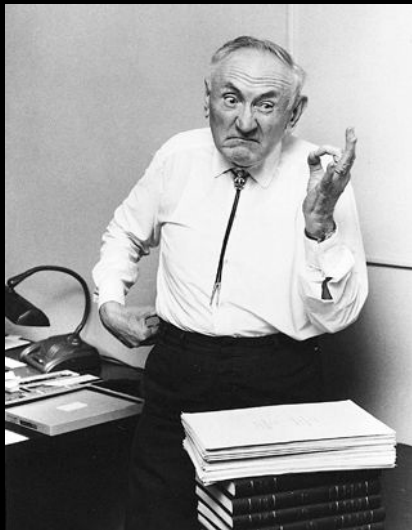




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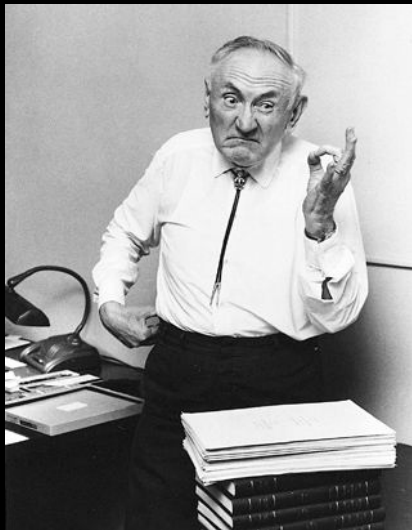
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- ▶ Right now it's really just a name for a phenomenon we don't understand yet.



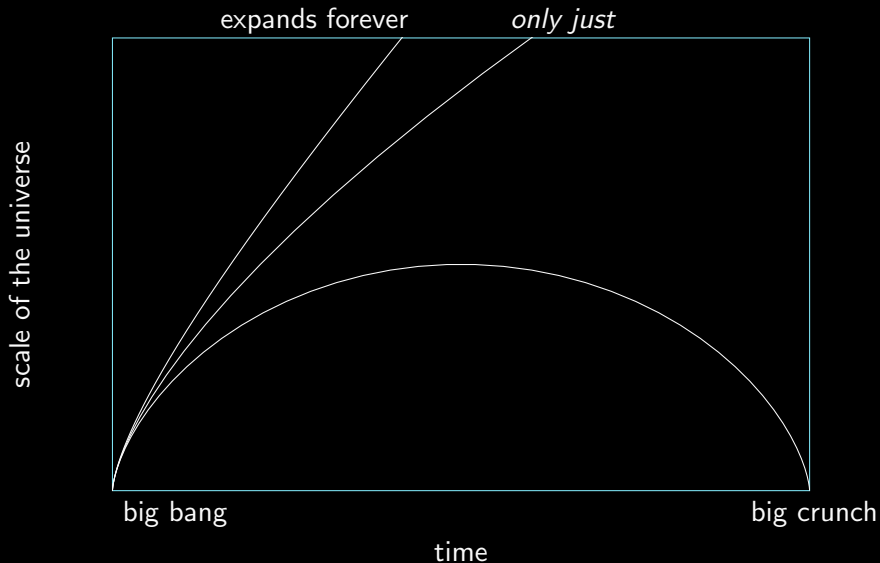
# Dark Energy

## The Problem



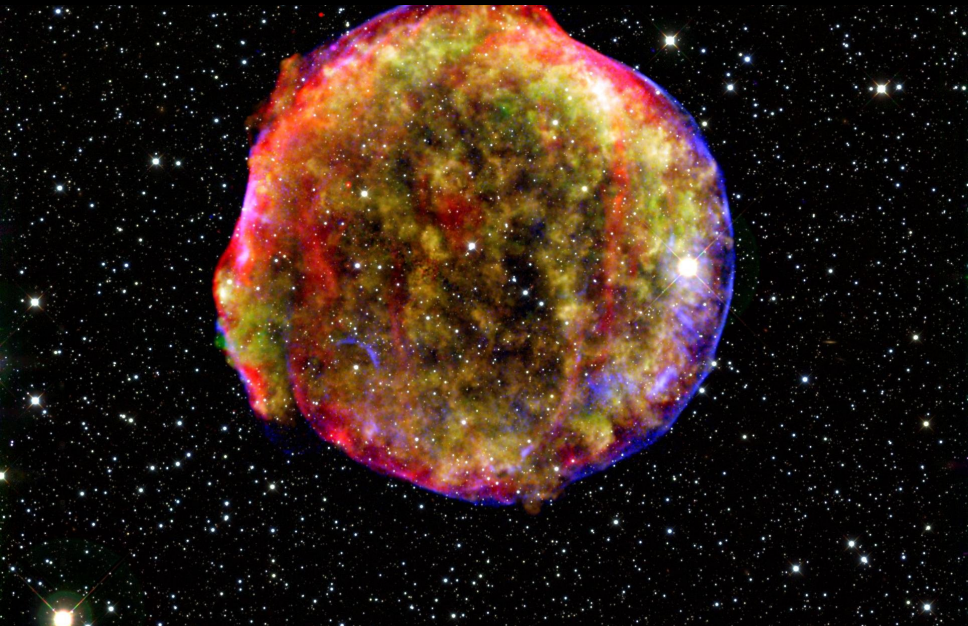
# The evolution of the universe

Three options:



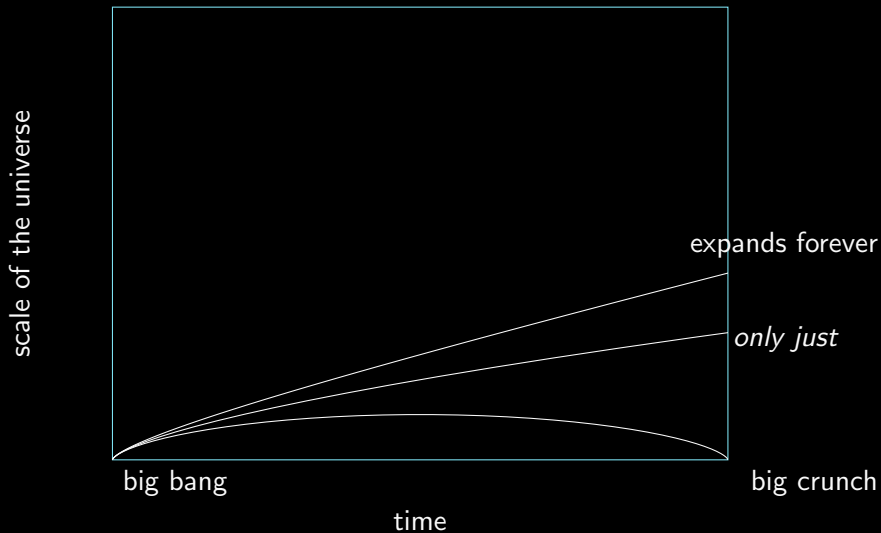
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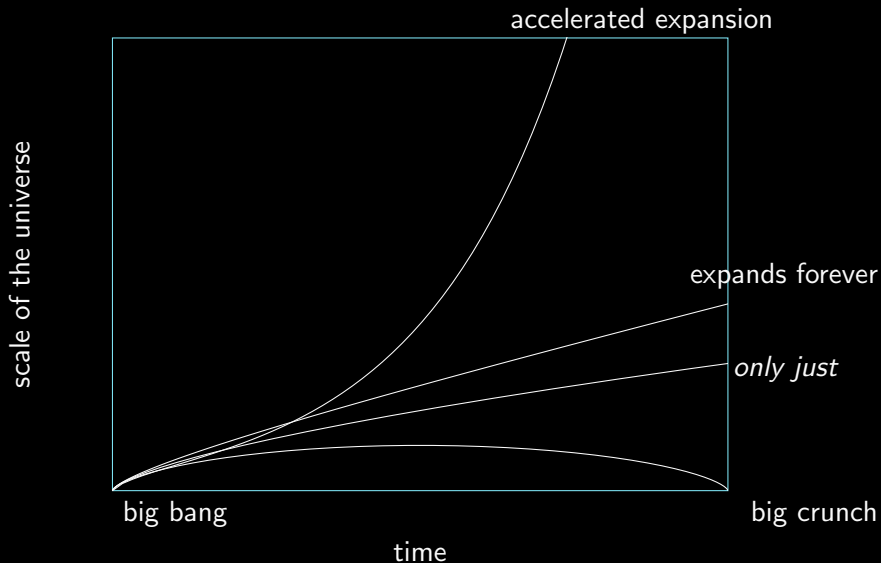
# The evolution of the universe

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# The evolution of the universe

None of the above:



# Dark Energy

## The Problem

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- ▶ One of the ways in which this could happen would be the presence of an energy permeating the universe, accelerating the expansion.
- ▶ We call this Dark Energy
- ▶ Like Dark Matter, it's really just a name for a phenomenon which we observe, and haven't been able to explain yet

# Dark Energy...

...or Cosmological Constant?

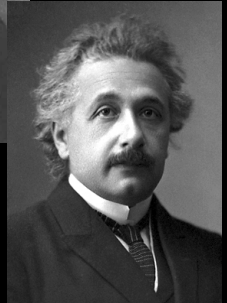
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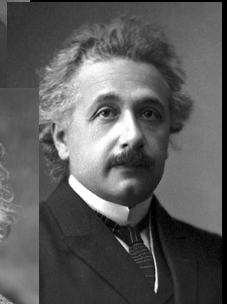
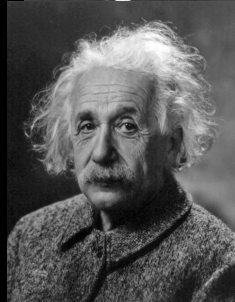
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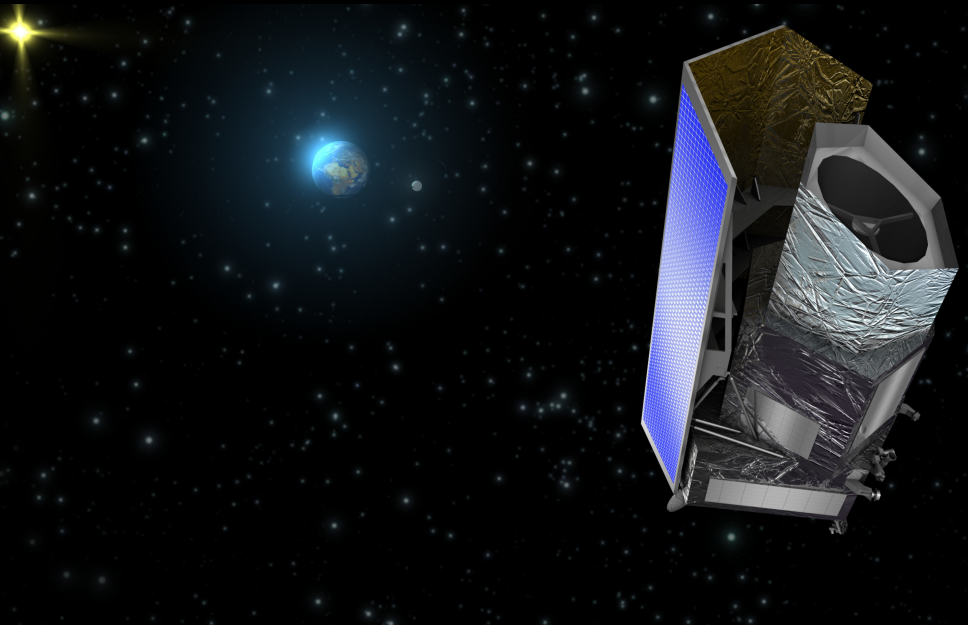
...or Cosmological Constant?

- ▶ Einstein's gravity can be repulsive at large scales.
- ▶ This is known as a cosmological constant.
- ▶ In modern terms, we think of it as a type of energy.



# The Euclid Satellite

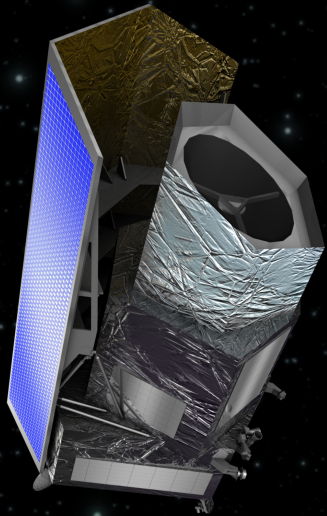
Exploring dark energy



# The Euclid Satellite

Exploring dark energy

► Launch date  $\approx$  2020

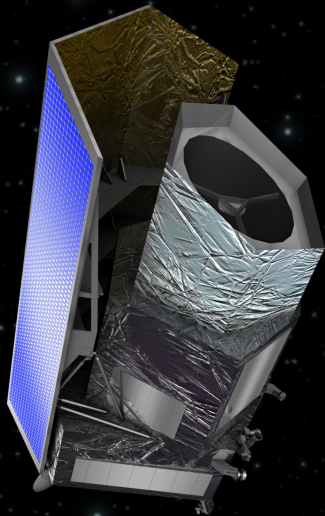




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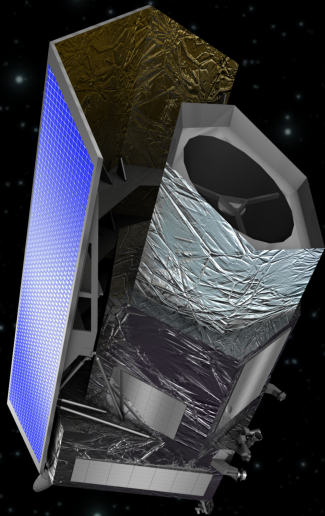
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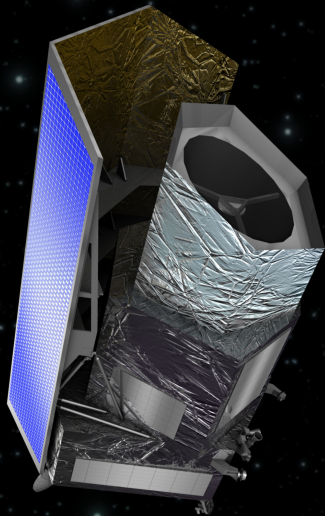
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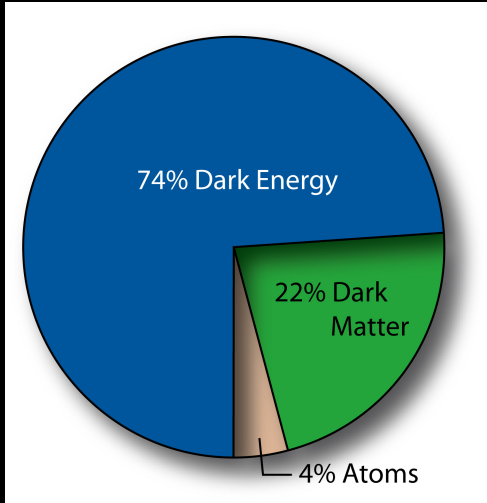
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Exploring dark energy

- ▶ Launch date  $\approx$  2020
- ▶ Build up an accurate 4D picture of the visible universe
- ▶ By measuring cosmic shear around galaxies and clusters, will be able to build up a map of Dark Matter
- ▶ Will see any variation in dark energy across space and time, enabling us to build a better understanding of it.

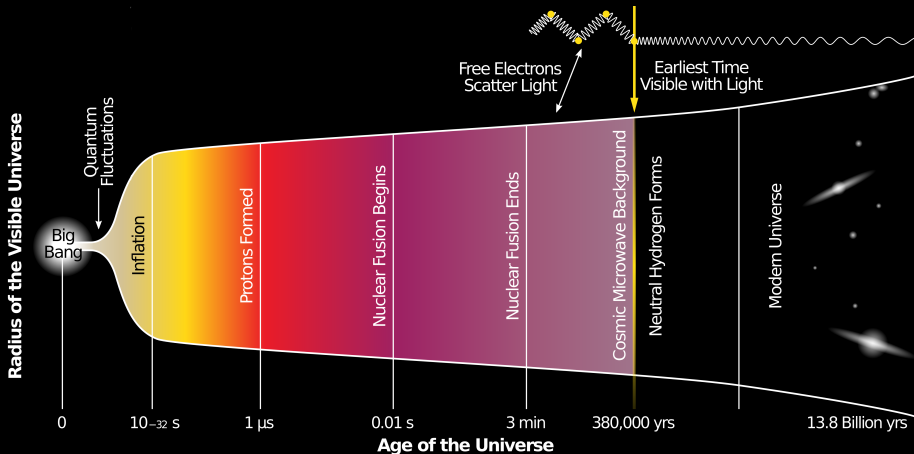


# The Composition of the Universe



# The beginning of time

$t = 0$



# Inflation

A convenient explanation

1. The CMB is far too uniform

# Inflation

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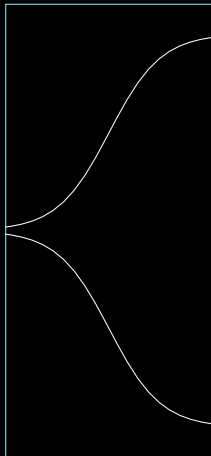
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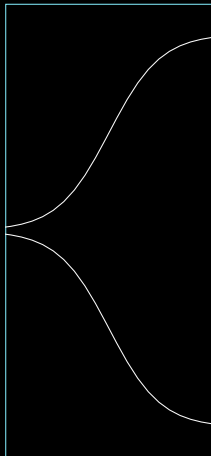


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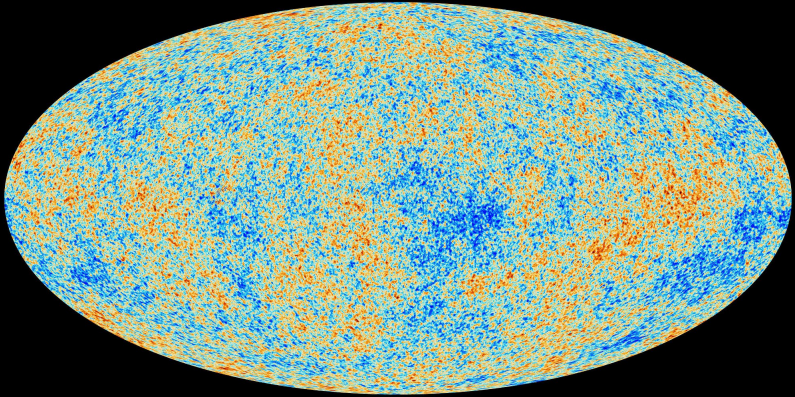
1. The CMB is far too uniform
2. It looks like it expanded incredibly rapidly “long before” the surface of last scattering.
3. We call this accelerated expansion “Inflation”

scale of the universe



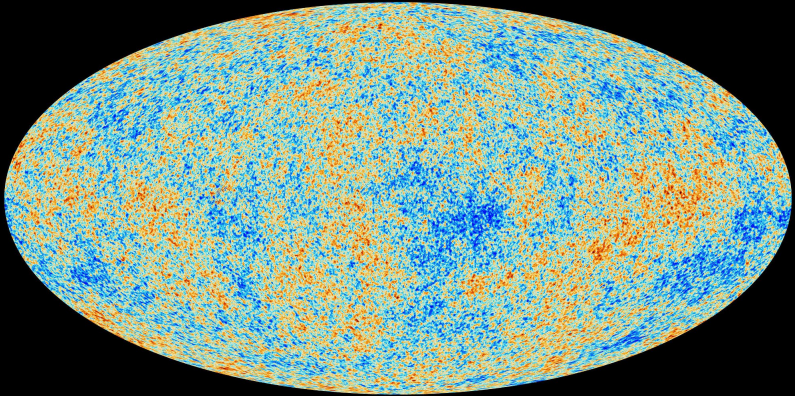
time

# Anisotropies in the CMB



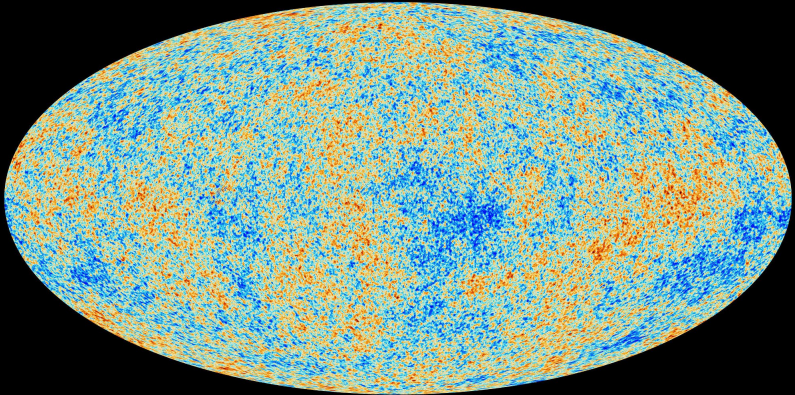
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- ▶ The CMB has a temperature of  $2.7260 \pm 0.0013$  K:
- ▶ The *Theory of Inflation* predicts these inhomogeneities,
- ▶ Which are created by *quantum fluctuations*.
- ▶ These are the seeds of structure formation in the universe.

# The dawn of a new era in astronomy

The rise of the mega-telescope

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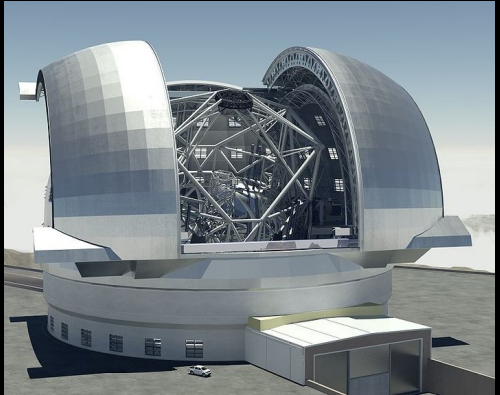
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- ▶ Cost: €1.5bn
- ▶ Testing GR, DE, DM, Magnetism...
- ▶ Imaging Hydrogen through the universe will build a 3D picture of the first ripples of structure.

# European Extremely Large Telescope (E-ELT)

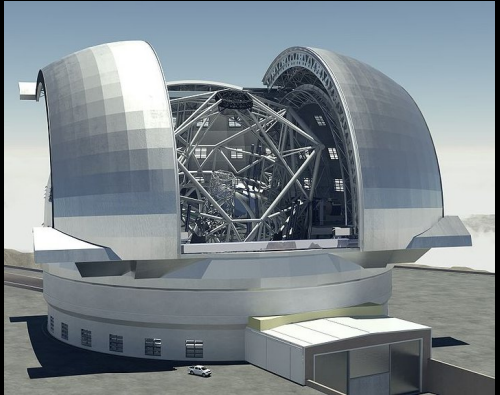
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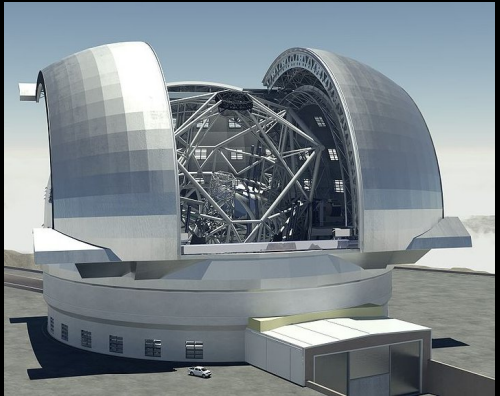
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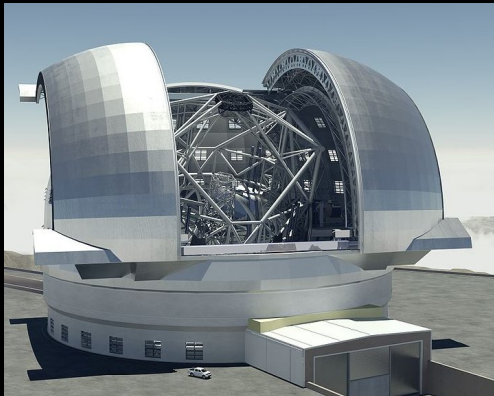
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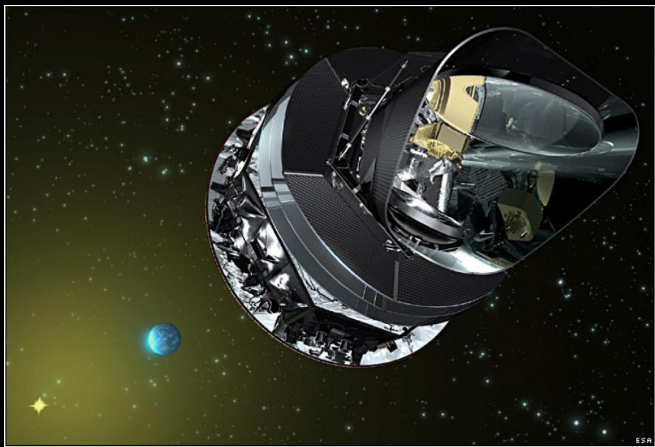
- ▶ Cost  $\approx$  €1.3bn
- ▶ Search for extrasolar planets.
- ▶ Over a 20 year baseline, it will be sufficiently accurate to watch the universe expand.





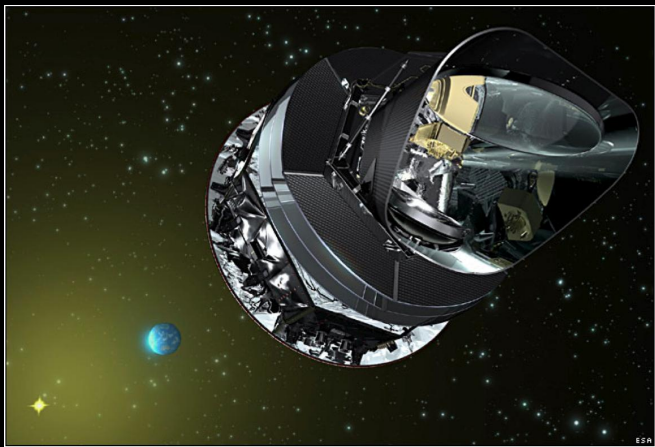
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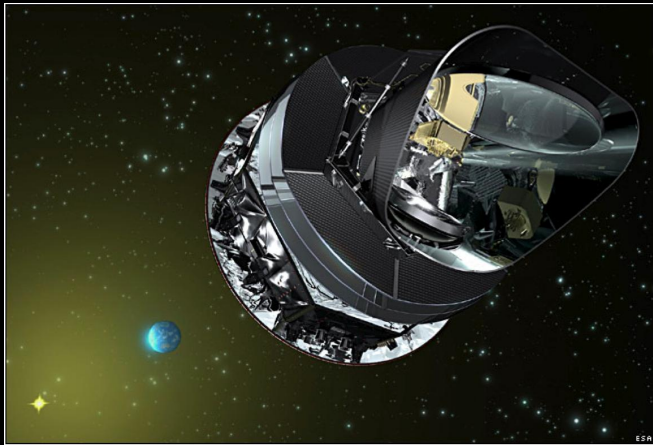
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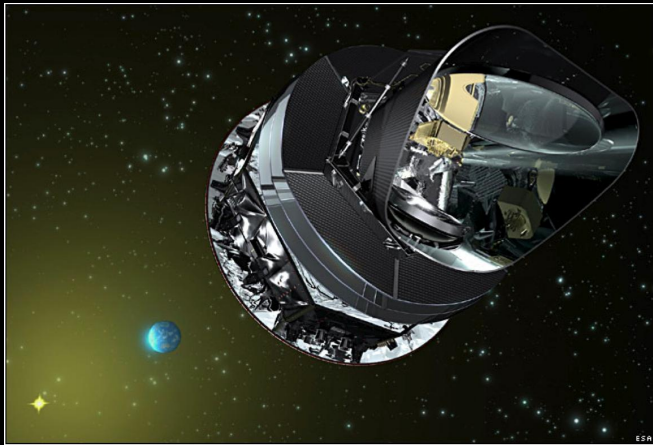
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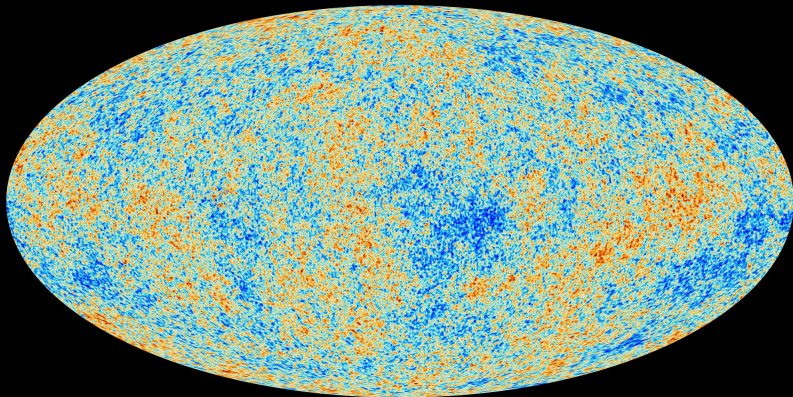
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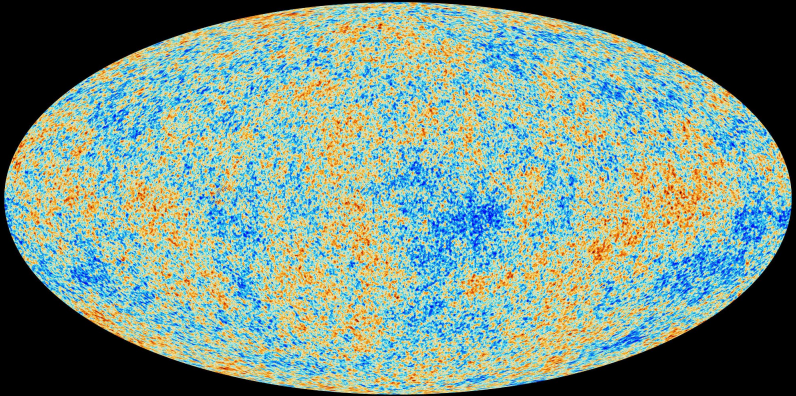


- ▶ Next-generation CMB mapping: polarisation
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- ▶ Providing essential data for accurate exploration of cosmological theories

# Summary

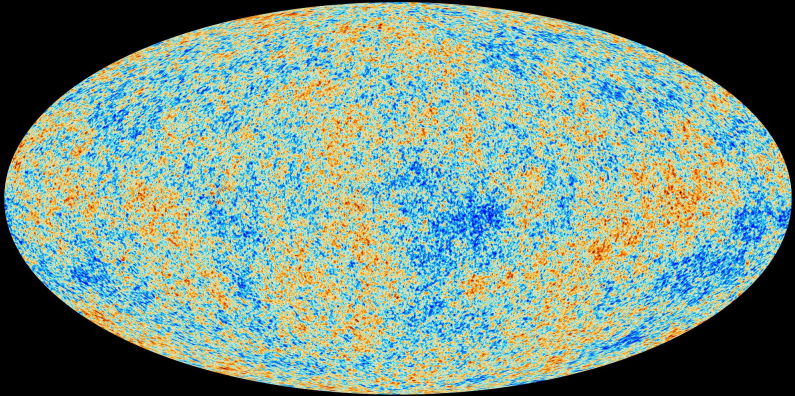


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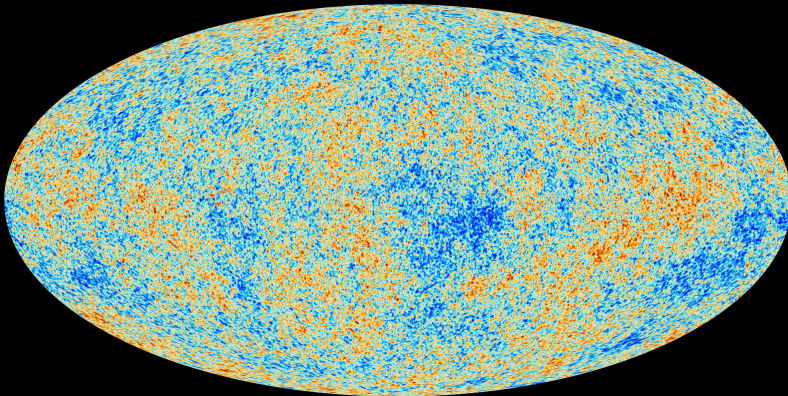
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- ▶ There are many unsolved theoretical problems,
- ▶ And a wealth of data to explore them with.



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